

Lab 1

Daniel Mehta #n01753264

Revision:

Control Structures

Create a loop that prints numbers from 1 to 10.

```
In [17]: for x in range(10):
        print(x+1)

1
2
3
4
5
6
7
8
9
10
```

Create a loop that prints even numbers between 1 to 10.

```
In [22]: for x in range(10):
        x+=1
        if (x % 2) == 0:
            print(x)

2
4
6
8
10
```

Use an if-else structure to check if a number is odd or even

```
In [31]: num = float(input("Enter Number: "))
        if (num % 2) == 0:
            print(f"{num} is even")
        else:
            print(f"{num} is odd")

4.0 is even
```

Functions

Write a function that takes a number as an argument and returns its square.

```
In [38]: def square(num):
        return num * num

square(5)

Out[38]: 25
```

Write a function that takes a string as an argument and returns its reverse.

```
In [41]: def reverse_string(string):
        return string[::-1]

reverse_string("Hello World!")

Out[41]: '!dlrow olleH'
```

Exercises:

Write a Python function named second_largest that takes a list of integers and returns the second-largest number in the list. Assume the list has at least two distinct elements.

```
In [52]: def second_largest(list):
        largest = second_largest = float('-inf')

        for n in list:
            if n > largest:
                second_largest = largest
                largest = n
            elif n > second_largest and largest != n:
                second_largest = n
        return second_largest

second_largest([5,2,8,1,7])

Out[52]: 7
```

Write a function named accumulate_squares that takes an integer n and returns the sum of the squares of all numbers from 1 to n that are not divisible by 3.

```
In [56]: def accumulate_squares(n):
        return sum(i**2 for i in range(1, n+1 ) if i% 3 != 0)

accumulate_squares(5)

Out[56]: 46

In [ ]:
```